One-Pager

FieldSignal

Always-on clinical triage and outbreak surveillance for disconnected field hospitals — running entirely on the Dell Pro Max with GB10.

Working name. Pick something a procurement officer can say out loud without explaining the joke.

The problem

In 1999, in South Sudan, a six-month delay in responding to a relapsing fever outbreak killed more than 2,000 people. WHO built EWARN in response. Twenty-seven years later, that system still runs on paper forms, manual case-definition matching, and a human counting cases toward a threshold.

Meanwhile, the most frequent request on MSF's telemedicine platform is for radiologists. Specialist reads take days. Complex cases wait for a weekly video call that sometimes can't connect at all.

Both problems have the same shape: **the clinical signal already exists at the point of care, and nobody can act on it fast enough.**

The product

A clinician works normally — dictates a consultation in their own language, uploads an X-ray from the portable unit, writes a note. Everything lands in a watched folder on the box.

FieldSignal runs continuously, unprompted, and:

Step	What happens
Listen	Speech-to-English translation in one pass. Source audio, native transcript, and English kept side by side for clinician verification.
Look	Medical vision model scores the film, producing a calibrated abnormality score rather than prose.
Structure	Maps each presentation onto formal WHO syndromic case definitions — the step that currently requires a trained surveillance officer.
Remember	Links the patient into a longitudinal graph, resolving returning patients who arrive with no ID and inconsistent name spellings.
Watch	Detects clusters across patients, across days, against real background noise.
Escalate	Threshold trips → clinician on shift is alerted in minutes, with the linked cases and the agent's reasoning.
Gate	A human approves before anything leaves. What crosses the wire is aggregate counts: *"eleven cases of acute watery diarrhoea, sector 4, past 72 hours, rising."* Never a record.

WHO already ships a physical product called "EWARS in a box" for surveillance in settings without reliable internet or power. This is the version that thinks.

Why it cannot run in the cloud

Not a preference. Three structural reasons:

1. **The data can get patients killed.** MSF's own data policy classifies as highest-risk any record implying criminal conduct or exposing patients to serious risk including death — explicitly naming violence-related data such as gunshot wounds, sexual violence records, and data from detention settings. In a conflict zone the threat model is not a compliance auditor; it's a combatant with legal process. No BAA addresses that.

2. **One remote call breaks the whole chain.** The pipeline is multimodal — speech, vision, retrieval, reasoning. If any single stage is an API call, the audio, the image, and the note all have to leave. Multi-model is what makes local-first load-bearing.
3. **Cloud models are a moving target.** Providers deprecate and silently update. Clinical triage behaviour changing underneath you without revalidation is unauditable. A frozen local model version inside the NemoClaw sandbox — default-deny, allowlisted endpoints, sandboxed writes — is reproducible.

The architecture's punchline: the most valuable thing the system produces is also the only thing safe to transmit. Privacy and usefulness point the same direction.

Why this machine

- **Always-on forbids model swapping.** Four models resident simultaneously — ASR, vision, embeddings, reasoning. If the heartbeat pages models in and out, load time dominates and the loop stops being continuous. Co-residency is a memory-capacity requirement; 128GB is what buys it.
- **Unified memory for imaging.** DICOM arrays are large; no PCIe copy between pipeline stages.
- **Two workloads at once.** Background surveillance sweeps while a clinician works in the foreground. Independent review of this exact box found sustained background agent loops with no noticeable foreground impact.
- **Power budget.** Field sites run on generators and solar. A deskside box deploys where a rack does not.

Why always-on

Four behaviours impossible without persistence — only the first is the demo:

1. **Outbreak surveillance** — inherently requires watching across days.
2. **Overnight backlog** — the day's films and notes process while nobody's there; the morning handover report is already written.
3. **Opportunistic sync** — waits for the connectivity window, pushes the prioritised digest. Nobody has to be awake.
4. **Longitudinal linkage** — returning patients resolve into the graph over time.

It's not a cron job: the agent chooses what to examine, calls tools, reasons about coincidence versus cluster, and drafts the clinical language. The trace panel in the demo is the evidence.

Who pays

Tier	Buyer	Why
Lighthouse	CSR / accessibility budgets at BigTech, hospital networks, health systems	Fast yes, low friction, generates the reference deployment
Scale	Remote industrial medicine — rigs, mines, cruise lines, expedition and research stations	Duty-of-care obligations, real medical budgets, identical technical constraints
For Dell and NVIDIA	Hardware pull-through	Every clinic, camp, rig, and ship that deploys this is one more GB10 sold into a segment that currently buys nothing. The software is the reason the box gets purchased.

Safety posture

It triages and prioritises. It never diagnoses.

- Every output is a draft for a clinician; every escalation path has a human gate.
- Alerts fire on structured fields a human can inspect — never on raw translated free text, so a hallucinated clause cannot manufacture an outbreak.
- Thresholds are calibrated per setting, which is already WHO's stated requirement for computer-aided detection.
- Model versions frozen and auditable inside the sandbox.

- Goal is shorter time-to-investigation, not automated outbreak declaration.

The demo — 5 minutes

0:00 Walk to the box. **Unplug the ethernet.** Hold it up, set it on the table in front of the judges. *"Everything you're about to see runs with that unplugged."*

0:10 South Sudan, 1999. Two thousand deaths. The system built in response is still paper.

0:35 The gap, one slide, one fact: MSF's most-requested telemedicine specialty is radiology; reads take days.

1:10 The demo, on the box, not slides:

- Three consultations dropped into the watched folder on camera — audio, X-ray, note.
- Agent picks them up unprompted. **Nobody touches the keyboard.**
- Trace panel: translates → scores the film → retrieves the case definition → queries the graph → reasons about signal versus noise.
- Alert fires. Clinician view with linked cases and reasoning. Approve.
- Sync window opens. **On-screen counter:** **`2.7 GB on box · 340 bytes transmitted`.** Cut to the patient records still sitting there.

3:30 Why this machine — four models resident, unified memory, concurrent workloads, deskside power. Then: *"every deployment is one more of these boxes."*

4:20 Close: *"It triages, it never diagnoses. And this can't be built on cloud inference — not because we'd rather not, but because cloud inference requires uploading the exact thing we're protecting. The cable stays out."*

Choreography

- **Record the 3-minute video by 5pm**, not 6:25. It's the artifact that survives if the live demo dies.
- **Exactly one thing is unscripted live:** dropping the files in. Models warm, graph pre-populated with two weeks of ordinary consultations so the cluster trips on three new arrivals against real noise.

- **Recorded fallback cued on the second monitor.** If the box hiccups: *"I'll show you the recording"* — and keep talking.
- **Demo language: French or Arabic.** Core MSF operating languages, well-supported by the ASR. Do not pick something exotic and produce garbage on stage.
- **Put the box on the table**, not behind the podium.

Rubric alignment

Criterion	Weight	How we score
Local-first + always-on	30%	Heartbeat runs unattended across days; privacy is structural, not preferential; proven live with the cable out
Business value	30%	Named buyers at three tiers, automates a WHO-standardised workflow that exists today, hardware pull-through for the sponsors
Demo + pitch	30%	One spine, one irreducible piece of theatre, opening line with a body count, closing line that makes the architecture inevitable
Technical execution	10%	Multi-model justified by the pipeline; only one unscripted moment on stage; recorded fallback

Q&A prep

- **"Isn't this a cron job?"** → It chooses what to examine, calls tools, reasons about coincidence versus cluster, drafts clinical language. Point at the trace.
- **"What about false alarms?"** → Human gate exists for exactly that. Per-setting threshold calibration is already WHO's requirement for CAD. We shorten time-to-investigation; we don't declare outbreaks.
- **"MSF has internet in most projects."** → Correct, and not our argument. Ours is data sensitivity plus specialist latency plus intermittency. Records implying violence can't leave regardless of bandwidth.

- **“Is this a medical device?”** → No, and designed not to be. Triage and prioritisation, human in the loop on every path.
- **“Who pays?”** → See the three tiers above.

Explicitly out of scope

Named deliberately — this is discipline, not incompleteness.

Radiologist-grade reporting · multilingual ASR beyond the demo language · per-language fine-tuning on-box · EMR and DHIS2 integration · deterioration monitoring for admitted patients · multi-site federation.

Sources worth citing in the deck

- WHO EWARS / EWARN, including “EWARS in a box” for settings without reliable internet or electricity — who.int/emergencies/surveillance
- South Sudan 1999 relapsing fever delay, >2,000 deaths — CDC *Emerging Infectious Diseases* 23(13), 2017
- WHO recommendation of computer-aided detection for TB screening since 2021, with per-population threshold calibration
- MSF telemedicine: radiology as most-requested specialty — doctorswithoutborders.org
- MSF data-sharing policy, sensitive-data classification — *PLOS Medicine*, Sheather et al.
- Dell Pro Max with GB10 sustained background agent loops — *Computer Weekly* review

PRD

FieldSignal — Technical Execution PRD

Version: 1.0 · Hackathon build · Dell x NVIDIA Local AI on Dell Pro Max with GB10 **Timebox:** Demo video + project submission by 18:30. Pitch deck by 19:00. Code freeze at video submission. **Team:** 3–4 builders, roster locked at kickoff.

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1. Product overview

One-line: An always-on clinical agent for disconnected field hospitals that turns every consultation into outbreak surveillance, with no patient data leaving the box.

Problem: WHO's EWARN outbreak-surveillance workflow in humanitarian settings still runs on paper forms and manual threshold counting. Separately, specialist reads (radiology especially) take days. The clinical signal exists at the point of care; nobody can act on it fast enough.

Primary user: A clinical officer at a district-level field hospital — not a radiologist, not an epidemiologist, running 40+ consultations a day, with intermittent connectivity and no specialist on site.

Secondary user: The project medical lead, who approves what escalates off-site.

Why this hardware: Four models must stay resident simultaneously for the heartbeat to be continuous. That is a memory-capacity requirement that 128GB unified memory satisfies and a 24GB consumer GPU does not.

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2. Goals & non-goals

Goals

#	Goal	Demo-verifiable success criterion
G1	Agent acts without being prompted	Files dropped in watched folder are processed with zero keyboard input
G2	Multimodal pipeline runs fully local	`nvidia-smi` / model list shows 4 resident models; egress policy blocks all non-allowlisted hosts
G3	Cluster detection across time	Alert fires on Nth case matching a syndromic definition within a window, against pre-populated background
G4	Human-gated, minimal egress	Approve action emits a JSON payload under 1KB containing counts only, no identifiers
G5	Visible agent reasoning	Trace panel shows tool calls in sequence during processing

Non-goals (explicit — do not build today)

- DICOM parsing. Use PNG/JPEG.
- Radiologist-grade report generation.
- More than one ASR source language.
- Per-language fine-tuning.
- EMR / DHIS2 / OpenMRS integration.

- Authentication, multi-user, RBAC.
- Deterioration monitoring for admitted patients.
- Multi-site federation.
- Any mobile client.

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3. User stories

P0 — must have, these are the demo

- As a clinical officer, I want my dictated consultation transcribed and translated to English so a colleague or the system can act on it.
- As a clinical officer, I want an uploaded chest X-ray scored automatically so I know which films need escalation first.
- As a clinical officer, I want to be told when my recent patients form a cluster, so I don't have to notice it myself across a week of notes.
- As a project medical lead, I want to review the agent's reasoning and approve before anything is transmitted.

P1 — should have, build only if P0 is green by 15:00

- As a clinical officer, I want to see the original audio, the native transcript and the English translation side by side so I can verify the translation.
- As a project medical lead, I want to see exactly how many bytes left the box.

P2 — nice to have, likely cut

- Returning-patient resolution across name-spelling variants.
- Overnight backlog batch summary.

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4. System architecture

```
/data/inbox/  ──watcher──▶  jobs (SQLite)  ──▶  pipeline workers
                            │
```

```
ASR/translate   vision scorer   case-def RAG   graph writer
                     |
            heartbeat loop (agent)
                     |
      tools: query_graph, get_case_def,
            score_film, raise_alert
                     |
        alert ──→ UI review ──→ approve
                       |
           egress (allowlisted, counts only)
```

Component responsibilities

Component	Responsibility	Tech
Watcher	Detect new files in `/data/inbox`, enqueue jobs	Python `watchdog` (inotify)
Job store	Durable queue + processed artifacts	SQLite
ASR worker	Speech → English translation, one pass	Whisper large-v3 via local serving
Vision worker	Chest film → abnormality score + findings	MedGemma (4B variant) via Ollama
Case-def RAG	Retrieve matching WHO syndromic case definition	Local embedding model + SQLite vector table
Graph	Patient / visit / syndrome nodes and edges	SQLite (nodes + edges tables). **Not** Neo4j — no time for a new service.

Agent	Heartbeat loop, tool calls, cluster reasoning, alert drafting	OpenClaw harness under NemoClaw / OpenShell
UI	Inbox view, trace panel, alert review, approve, byte counter	FastAPI + server-rendered HTML, no SPA
Egress	Emit approved aggregate payload to the single allowlisted endpoint	HTTP POST to a mock receiver on a second machine

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5. Devbox specification and setup

Hardware: Dell Pro Max with GB10 — Grace Blackwell, 128GB coherent unified memory, Gen4 NVMe, DGX OS (Ubuntu 24.04 base). One box per team, plus monitor and keyboard/mouse. Team codes on personal laptops; **the demo must run on the box**.

Required stack per Rule 02: NemoClaw, OpenClaw, or OpenShell. All inference local. No remote LLM/API calls in the agent's runtime path.

Setup sequence (owner: Infra, target complete by 11:30)

1. Verify Docker 28+ and the NVIDIA container runtime are present.
2. `nemoclaw onboard` — provisions the OpenShell sandbox, configures inference routing, applies filesystem and network policy from first boot.
3. Configure Ollama to serve on all interfaces via systemd drop-in.
4. **Load models from the USB/external drive.** Do not download at the venue — Wi-Fi will not support it.
5. Preload and pin weights so cold-start latency doesn't dominate the heartbeat.
6. Set the egress allowlist to exactly one host (the mock receiver). Verify a non-allowlisted request is denied.

Model memory budget

Sum must leave ≥20% headroom in 128GB so the heartbeat and foreground work don't contend. Verify actual resident sizes with `ollama ps` after load and adjust before committing.

Role	Model	Budget
Agent reasoning	Nemotron 3 Super — **quantized or smaller variant**	~30–50 GB
Medical vision	MedGemma 4B	~9 GB
ASR + translation	Whisper large-v3	~10 GB
Embeddings	Small local embedding model	~1 GB
Total target		**≤ 70 GB**

Decision: the full 87GB Nemotron 3 Super 120B is the flashy choice and the wrong one. It starves the other three models, kills co-residency, and co-residency *is* the pitch. Use the quantized/smaller variant. If asked in Q&A: "the box can run the 120B — but the product needs four models hot at once, and that's the better use of the memory."

Critical config

- **Model keep-alive must be infinite.** If models unload between heartbeat cycles, load time dominates and the "always-on" claim collapses. Set `OLLAMA_KEEP_ALIVE=-1` and raise `OLLAMA_MAX_LOADED_MODELS` to at least 4.
- **Prove the egress policy on camera.** The OpenShell default-deny architecture (allowlisted endpoints, sandboxed writes) is your evidence for Rule 02 and for the "no cloud calls" criterion. Have a terminal ready showing a blocked outbound request.

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6. Functional requirements

ID	Requirement	Priority	Owner

F-01	Watcher detects new file in `/data/inbox` and enqueues within 2s	Must	Infra
F-02	Audio file → English translation, persisted with native transcript	Must	ML-A
F-03	Image file → abnormality score (0–100) + short findings text	Must	ML-B
F-04	Extracted presentation mapped to a WHO syndromic case definition	Must	ML-B
F-05	Case written to graph as patient/visit/syndrome nodes with timestamp and catchment	Must	Backend
F-06	Heartbeat runs every 30s, queries graph, evaluates thresholds	Must	Agent
F-07	Agent raises alert with linked case IDs and natural-language rationale	Must	Agent
F-08	UI shows live trace of agent tool calls in sequence	Must	Frontend
F-09	UI alert review with Approve / Dismiss	Must	Frontend
F-10	Approve emits allowlisted POST with aggregate counts only; byte size displayed	Must	Backend
F-11	UI shows audio + native transcript + English side by side	Should	Frontend
F-12	Returning-patient resolution across name variants	Nice	Backend

F-07 detail: alert logic

- **Trigger:** ≥3 cases matching the same syndromic case definition, same catchment, within a rolling 72h window.

- **Input to the agent:** structured fields only — syndrome code, timestamp, catchment, film score. **Never raw translated free text.** A hallucinated clause must not be able to manufacture an outbreak.
- **Agent output:** `{severity, syndrome, case_ids[], window, rationale_text}`.
- **Edge cases:** fewer than 3 cases → no alert. Cases spread across catchments → no alert (demonstrate this if time allows; it shows the thresholds are real). Duplicate file dropped twice → dedupe on content hash.

F-10 detail: egress payload


```
{"syndrome":"acute_watery_diarrhoea","catchment":"sector-4",
 "count":11,"window_hours":72,"trend":"rising","site_id":"FS-001"}
```


No names, no ages, no free text, no identifiers. Display `bytes_sent` in the UI next to `bytes_on_box`.

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7. Non-functional requirements

Area	Requirement
Latency	Single case end-to-end < 45s. Heartbeat cycle < 10s when no new cases.
Concurrency	Background heartbeat must not block foreground UI. Demonstrate both running simultaneously.
Locality	Zero non-allowlisted network calls during the entire demo. Enforced by OpenShell policy, not by convention.
Durability	State survives process restart — SQLite on disk, not in-memory. A crash 20 minutes before the pitch must not lose the pre-populated graph.
Auditability	Every agent tool call logged with timestamp and arguments. This log *is* the trace panel.

Reliability	Demo path must be idempotent and re-runnable in under 60s. You will run it more than once.

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8. Demo data requirements

Asset	Source	Rule 06 note
Chest X-rays	Public de-identified set (NIH ChestX-ray14, VinDr-CXR). **Load from drive.**	Must be cleared for use and declared in the writeup
Consultation audio	**Record yourselves** in French or Arabic	Self-authored, no clearance issue
Background graph	Synthetic — 2 weeks of ordinary consultations, scripted	Self-authored
Cluster cases	Synthetic — 3 cases designed to trip the threshold	Self-authored
Case definitions	WHO syndromic definitions, paraphrased into your own schema	Cite source

Pre-populating the background graph is not cheating — it's realistic, and it means cluster detection runs against real noise instead of firing on an empty table.

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9. Build schedule

Time	Milestone	Cut line

now → 11:30	Box provisioned, NemoClaw onboarded, 4 models loaded and pinned, egress policy verified	If models aren't hot by 11:30, drop to 3 models and cut the embedding/RAG step
11:30 → 13:30	Watcher + SQLite + graph schema + one worker end-to-end on a single file	—
13:30 → 15:00	All three workers wired; agent tool definitions; heartbeat loop firing	**15:00 gate:** if a case doesn't flow end-to-end, cut F-04 and hardcode syndrome mapping
15:00 → 16:30	UI: inbox, trace panel, alert review, approve, byte counter	**16:00 gate:** if trace panel isn't rendering, print tool calls to a terminal and screen-capture that instead — do not lose the trace
16:30 → 17:15	Pre-populate background graph. Full dry run, twice. Fix what breaks.	—
17:15 → 18:00	**Record the demo video.** Everything working, tight cuts.	Non-negotiable — start recording at 17:15 even if P1 items are unfinished
18:00 → 18:30	Writeup, GitHub URL, submit via BuilderBase	Submit at 18:15, not 18:29
18:30 → 19:00	Pitch deck	—

Workstream ownership (4 builders)

- **Infra:** box setup, NemoClaw/OpenShell, model loading, egress policy, deployment
- **ML-A:** ASR + translation worker
- **ML-B:** vision scorer + case-definition mapping

- **Backend/Agent:** graph schema, heartbeat, agent tools, alert logic
- **Frontend** (shared, whoever finishes first): UI and trace panel

If 3 builders: Infra absorbs Frontend, and ML-A takes case-definition mapping.

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10. Risks

Risk	Severity	Mitigation
Models not on drive; venue Wi-Fi can't download them	**Fatal**	Verify drive contents *before* touching anything else
Model swapping kills heartbeat continuity	High	`KEEP_ALIVE=-1`, verify with `ollama ps` under load
ASR output is garbage in chosen language	High	Demo in French or Arabic. Test with real recorded audio by 13:00, not 17:00
Vision model produces prose, not a score	Medium	Constrain output with a strict JSON schema prompt; parse and validate; fall back to a fixed score if unparseable
Trace panel eats the whole afternoon	Medium	It's 30% of the score — protect it. Terminal output is an acceptable fallback
Live demo dies on stage	Medium	Recorded video cued on second monitor. Practice the handoff line
Scope creep back to five features	High	Non-goals list in §2 is binding. Re-read it at every gate

11. Open questions to resolve by 11:30

1. Demo language — French or Arabic? (Drives ASR testing.)
2. Final Nemotron variant and actual resident size after quantization.
3. Product name (FieldSignal is a placeholder).
4. Where does the mock egress receiver live — a teammate's laptop on the venue network, or a second local port? (Laptop is more convincing on camera.)
5. Who reads the pitch? Decide now; that person should not be debugging at 17:00.

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12. Submission checklist

- **What have you built** — writeup covering value to companies, and **declare your stack** per Rule 06
- **Demo video file** — max 3 min, uploaded
- **Demo video URL** — optional
- **GitHub URL** — required
- Confirm all third-party content (X-ray datasets, models) is cleared and cited
- Confirm the demo runs on the box with no remote API calls in the runtime path